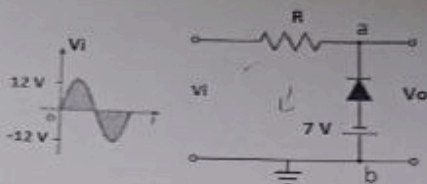
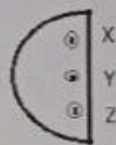


For the given circuit below, the reading of DMM if it is put between the points a and b for half cycle only is given as (Assume the diode is a practical Si diode)
 a. -7 V b. 5 V c. 19 V d. -5 V e. 12 V f. Not mentioned.



For the same circuit in Q9 above with $R = 2 \text{ K } \Omega$ but assume that the diode is ideal diode, the measured current through the resistor should be (Ignore the current sign)
 a. 2.5 mA b. 9.5 mA c. 6 mA d. 3.5 mA e. Not mentioned

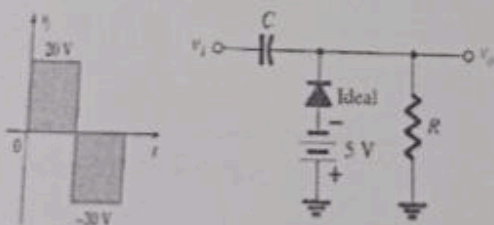
If the following figure represents the bottom view of FET discussed in the lab, then X, Y, and Z represent....., which are three terminals of FET.
 a. S, and D b. D, G, and S c. S, G, D d. D, S, G e. Not mentioned



In the BJT experiments, the current gain or amplification factor in common base BJT is defined as α and in common emitter BJT is defined as β

a. $\frac{I_C}{I_B}$ b. $\frac{I_C}{I_B} \cdot \frac{I_C}{I_E}$ c. $\frac{I_C}{I_E} \cdot \frac{I_C}{I_B}$ d. $\frac{I_C}{I_B} \cdot \frac{I_E}{I_E}$ e. Not mentioned

In the given clamper circuit using an ideal diode, the capacitor voltage will charge up to
 a. 15 V b. -5 V c. -15 V d. -35 V e. -35 V f. Not mentioned

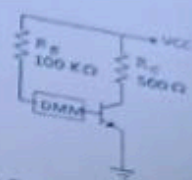


Compared to the BJT transistor, the MOSFET transistor is with
 a. Controlled device, higher input resistance.
 b. Controlled device, lower input resistance.
 c. Controlled device, high input resistance.
 d. Controlled device, lower input resistance.

For the FET transistor discussed in the lab, the characteristic curve of n-type channel is values of V_{GS}

a. Positive variable values b. Negative variable values c. Can not be determined.

16. For the figure shown below, the BJT configuration name is
 a. Common base b. Common collector c. Fixed bias configuration d. Not mentioned
17. For the same figure in Q16 above, assume that the BJT is a Silicon transistor for $V_{CC} = 10\text{ V}$ and $\beta = 100$, then the reading of DMM is
 a. $68\text{ }\mu\text{A}$ b. $39\text{ }\mu\text{A}$ c. 3.9 mA d. $93\text{ }\mu\text{A}$ e. 6.8 mA f. Not mentioned



For the given circuit shown in Figure 1 below, answer the following question from Q18- Q20.

- Q18. For an input voltage $V_s = 4\text{ V}$, the reading of DMM if it is between points A and B is approximately
 a. 2 mA
 b. 0.67 mA
 c. 2 mV
 d. 1.25 mA
 e. 1.2 mV
 f. Not mentioned.

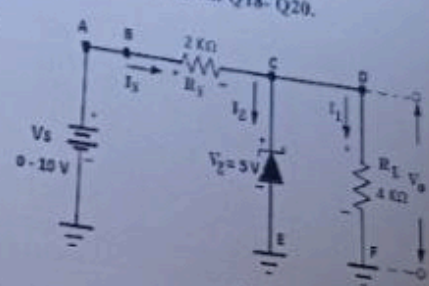


Figure 1

- Q19. For an input voltage ($V_s = 8\text{ V}$), the reading of DMM if it is put on the resistor R_2 is given as
 a. 5 V b. 1.5 V c. 1.5 mA d. 3 V e. 1.3 mA f. Not mentioned.
- Q20. For an input voltage ($V_s = 10\text{ V}$), the reading of DMM if it is put between point C and D is approximately
 a. 2.5 mA b. 2.5 V c. 0 V d. 5 V e. Not mentioned.

Q2. Using the following figure to answer the following questions from Q21 to Q25.

- Q2.1. Based on the experiments discussed in the lab, what is the name / objective of this circuit.

Answer: Half wave rectifier

